

ROOM GENERATOR

Overview

The **Room Generator** is a Unity tool that procedurally builds rooms based on customizable parameters. It can generate:

- **Floors** (with randomized tile prefabs)
- **Walls** (with optional doors, corners, halves)
- **Foundations** (support under the floor)

The system supports random prefab selection, multiple floors (stacked walls), and customizable grid sizes.

Inspector Breakdown

Room Dimensions

- **Grid X / Grid Y** – Number of tiles along the X (width) and Y (depth) axes.

Room Settings

- **Tile Size** – World size of each floor tile.
- **Foundation Size** – Size of foundation pieces under the floor.
- **Wall Size (X, Y)** – Dimensions of each wall prefab.
- **Tile Offset (X, Y)** – Adjusts floor tile placement offsets.
- **Wall Offset (X, Y)** – Adjusts wall placement offsets.
- **Door Pos (X, Y)** – Specifies door placement along X (front wall) and Y (side wall).
- **Floor Count** – Number of stacked wall layers (vertical levels).
- **Make Floor** – Toggle floor generation.
- **Make Wall** – Toggle wall generation.
- **Make Foundation** – Toggle foundation generation.
- **Not Rotation** – If enabled, tiles won't be randomly rotated.

Prefab Lists (Randomized)

- **Tiles** – Floor tiles.
- **Walls** – Standard wall pieces.
- **Wall Corners** – Special corner prefabs.
- **Wall Halves** – Shorter wall prefabs (used at corners).
- **Doors** – Door prefabs placed at specified positions.
- **Floor Foundation** – Foundation prefab used under the floor.

Buttons

- **Generate Room** – Builds a new room with the given settings.
- **Clear Room** – Deletes previously generated structures.